AWS HIGH AVAILABILITY LOAD BALANCING & BLUE-GREEN DEPLOYMENT

Using Amazon EC2, Application Load Balancer, Target Groups and Apache

Practical Project Documentation

AWS Region: Asia Pacific (Mumbai) — ap-south-1

Deployment model: Blue Environment (Version 1) and Green Environment (Version 2)

Prepared for project submission

1. Project Overview

This project demonstrates AWS Application Load Balancing and a manual Blue-Green deployment workflow. Two Ubuntu EC2 instances host separate versions of a simple Apache website. An internet-facing Application Load Balancer (ALB) receives HTTP requests and forwards them to target groups. The listener configuration can route traffic to Blue, Green, or distribute traffic between both target groups using weighted forwarding.

1.1 Objectives

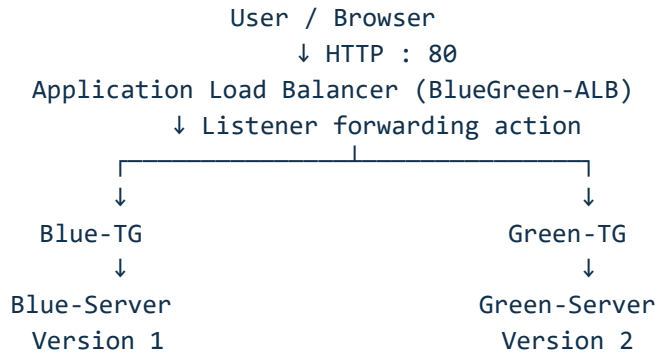
- Create a dedicated VPC with public subnets in two Availability Zones.
- Launch two EC2 instances to host separate application versions.
- Configure security groups for the ALB and EC2 instances.
- Create separate target groups and verify target health.
- Configure an internet-facing Application Load Balancer with an HTTP listener.
- Test Blue-to-Green traffic switching and rollback using the same ALB DNS name.
- Configure weighted forwarding with 50% weight for Blue and 50% for Green.

1.2 Project Scope

The project uses manual AWS Console configuration. It does not implement automatic scaling, a CI/CD pipeline, infrastructure as code, HTTPS/TLS, or application/database migration. The Blue-Green switch and weighted forwarding are controlled through the ALB listener rules.

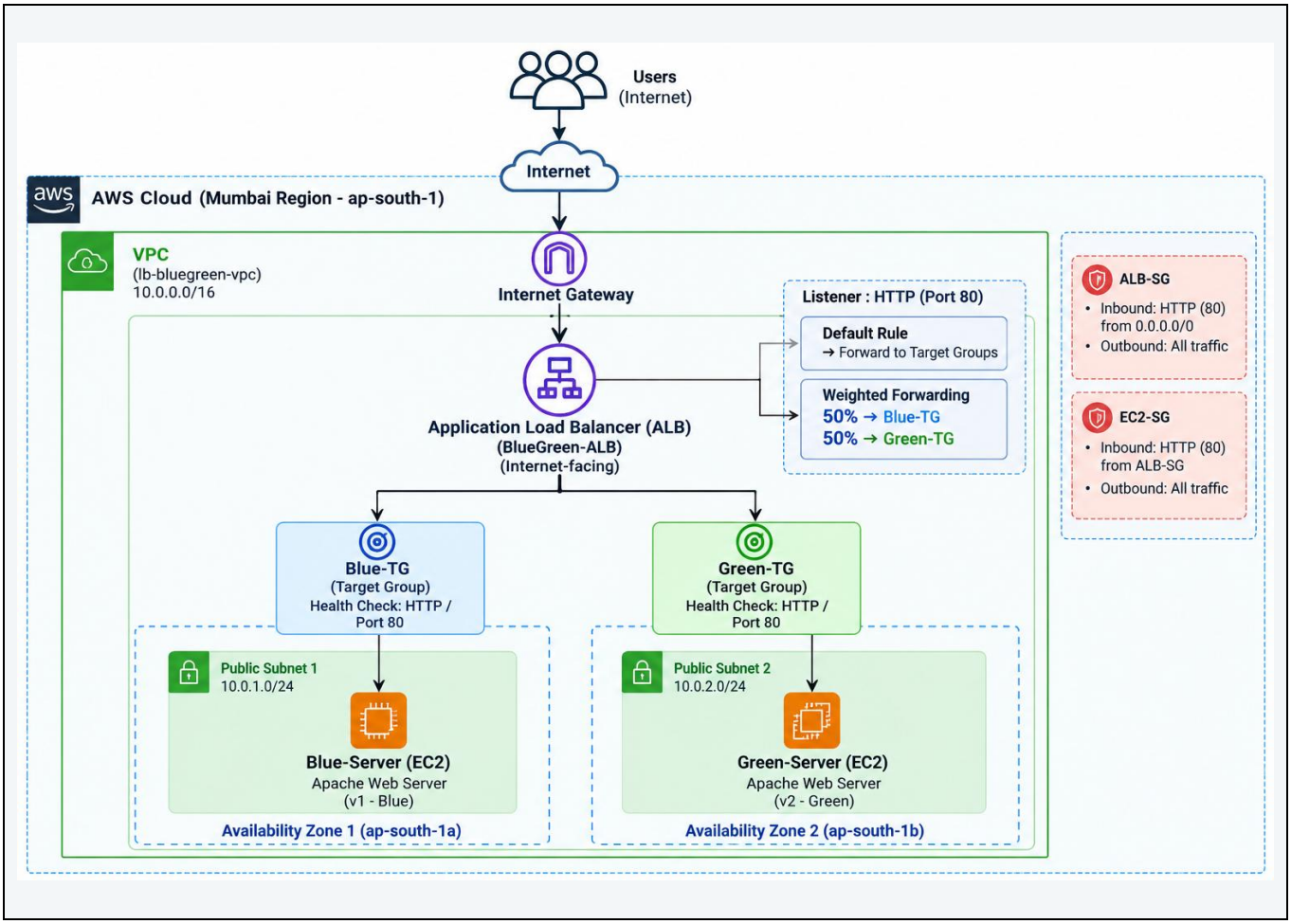
2. Architecture and Traffic Flow

The browser sends an HTTP request to the ALB DNS name. The ALB listener evaluates its forwarding action and sends the request to the selected target group. Each target group contains one EC2 instance running Apache.



The ALB was configured across two public subnets in different Availability Zones. The EC2 instances were placed in the project's public subnets for this learning setup. The EC2 security group allowed HTTP on port 80 from the ALB security group.

Figure 1. Project architecture / VPC overview.



3. AWS Resources and Configuration

Resource	Name / Configuration	Purpose
AWS Region	ap-south-1 (Mumbai)	Region used for all project resources.
VPC	lb-bluegreen-vpc; 10.0.0.0/16	Dedicated virtual network for project resources.
Public Subnets	2 public subnets in separate Availability Zones	Provide subnet mapping for the internet-facing ALB and EC2 instances.
ALB Security Group	ALB-SG; inbound HTTP TCP 80 from 0.0.0.0/0	Allows users to access the ALB over HTTP.
EC2 Security Group	EC2-SG; inbound HTTP TCP 80 from ALB-SG	Allows web traffic from the ALB to the EC2 web servers.
Blue EC2	Blue-Server; Ubuntu; Apache; Version 1	Hosts the Blue Environment website.
Green EC2	Green-Server; Ubuntu; Apache; Version 2	Hosts the Green Environment website.
Blue Target Group	Blue-TG; HTTP port 80; health path /	Registers Blue-Server.
Green Target Group	Green-TG; HTTP port 80; health path /	Registers Green-Server.
Load Balancer	BlueGreen-ALB; internet-facing; IPv4	Receives user requests and forwards them to target groups.
Listener	HTTP port 80	Routes traffic according to the configured default forwarding action.

Figure 2. VPC and subnet configuration.

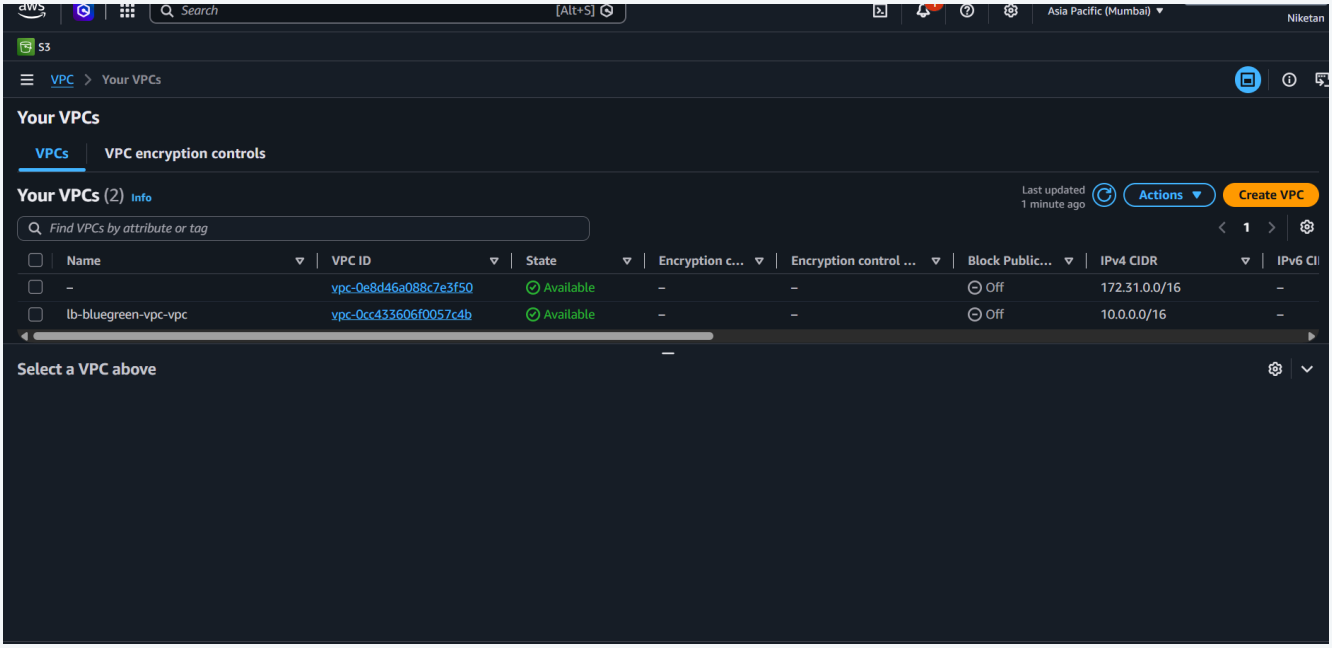
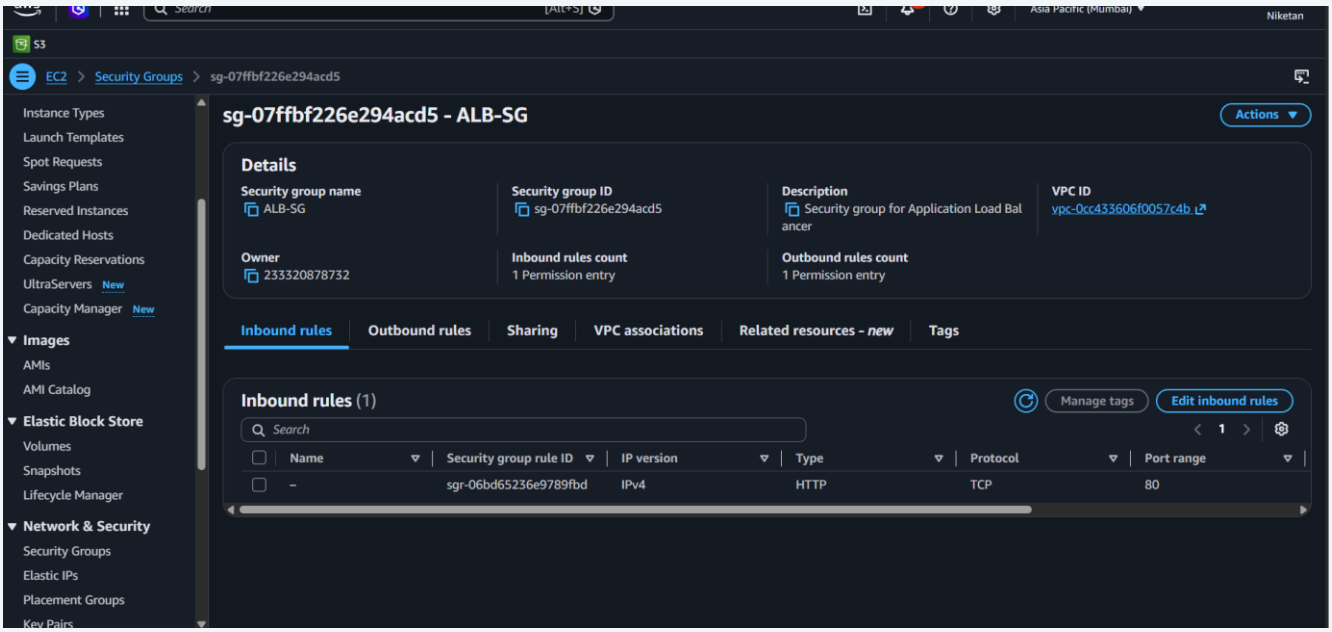


Figure 3. Security group rules.

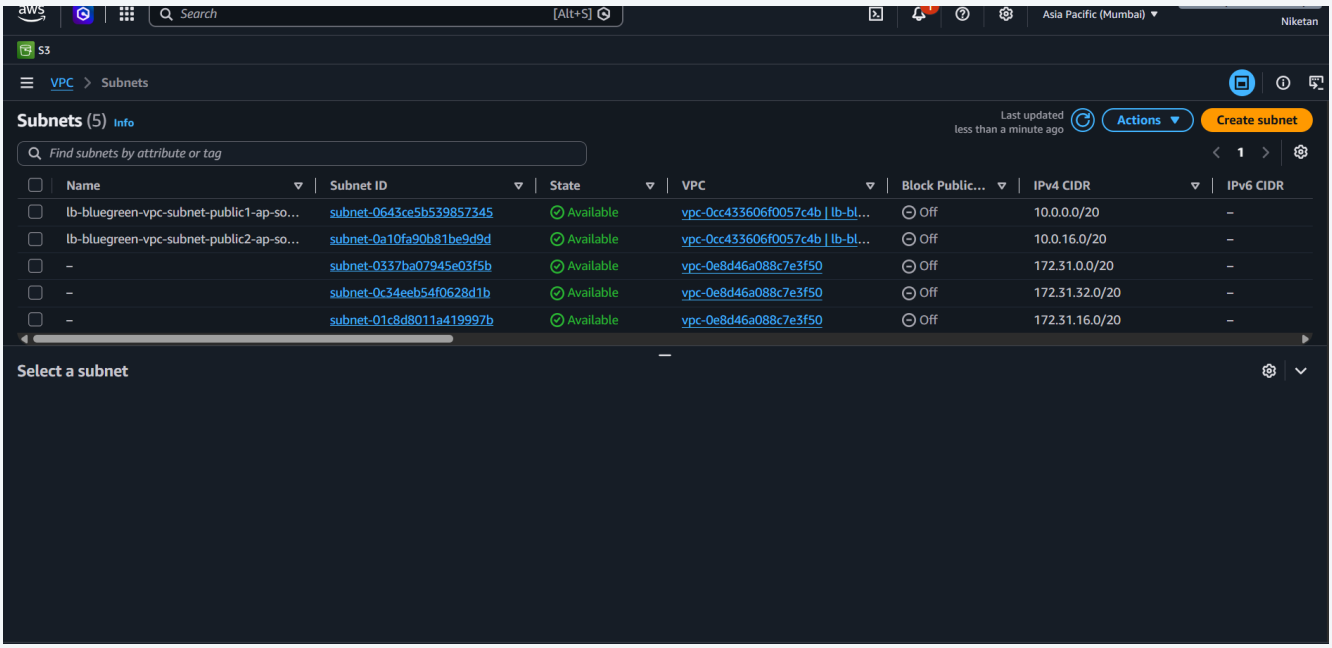


4. Implementation Steps

Step 1 — Create the VPC and Public Subnets

A custom VPC named lb-bluegreen-vpc was created with IPv4 CIDR 10.0.0.0/16. Two public subnets were configured in separate Availability Zones. NAT Gateways and private subnets were not created for this learning project. The public subnet route tables were configured for internet access through an Internet Gateway.

Figure 4. Created VPC and subnet resources.



Step 2 — Configure Security Groups

Two security groups were created. ALB-SG allowed inbound HTTP traffic on TCP port 80 from 0.0.0.0/0. EC2-SG allowed inbound HTTP traffic on TCP port 80 with ALB-SG as the source. This limits web traffic to requests coming through the load balancer. No public SSH rule was required for the demonstrated web traffic.

Figure 5. ALB-SG inbound rule.

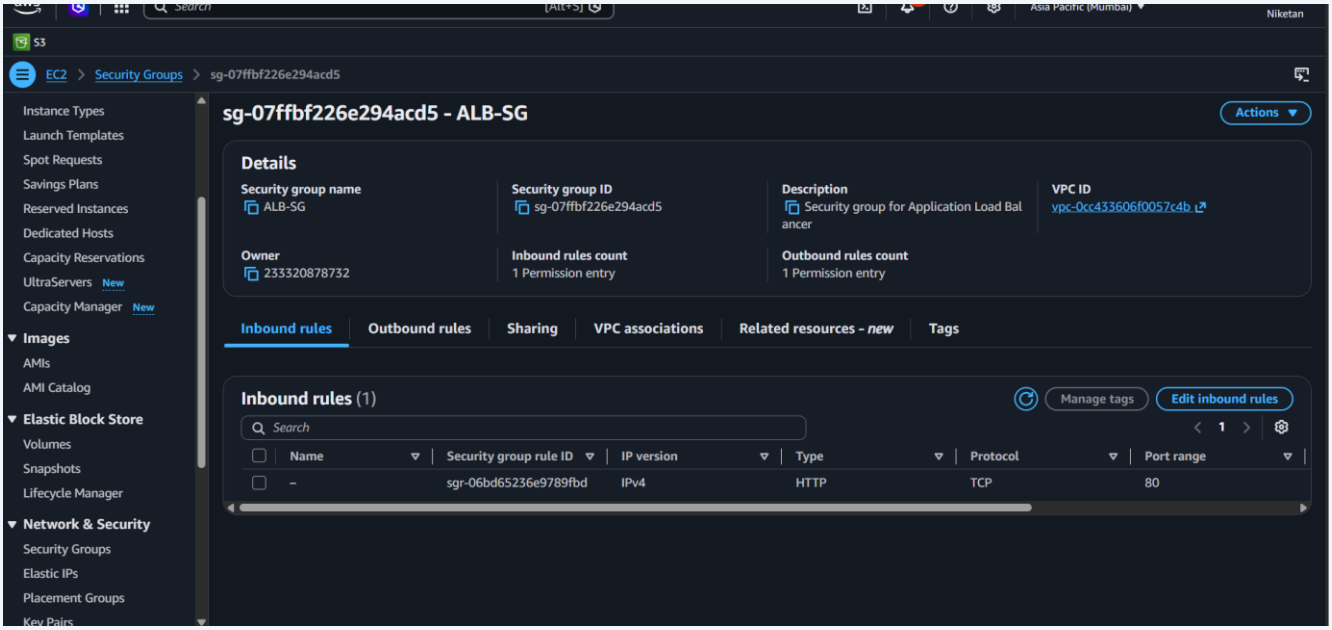
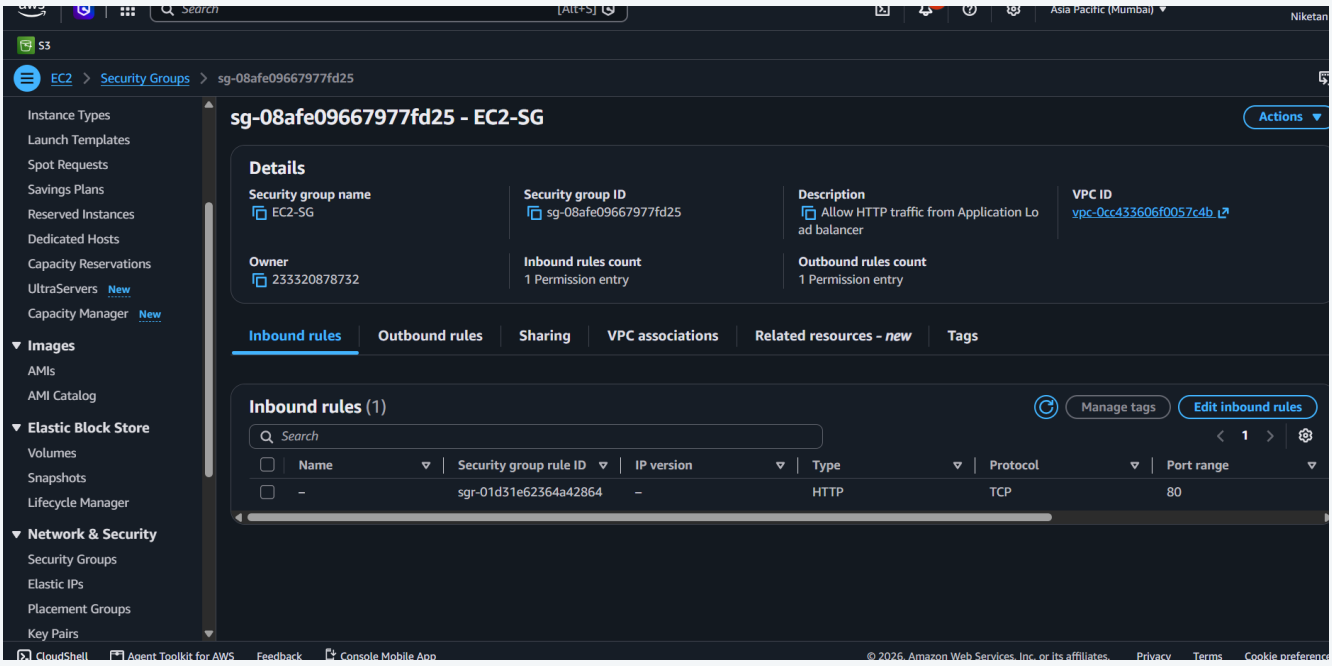


Figure 6. EC2-SG inbound rule.



Step 3 — Launch the Blue EC2 Instance

An Ubuntu Server EC2 instance named Blue-Server was launched in the project VPC. Apache was installed using instance User Data, and the default web page was replaced with a page displaying “BLUE ENVIRONMENT” and “Application Version 1”. The instance was associated with EC2-SG.

Step 4 — Launch the Green EC2 Instance

A second Ubuntu Server EC2 instance named Green-Server was launched in the other public subnet. Apache was installed using User Data, and its web page displayed “GREEN ENVIRONMENT” and “Application Version 2”. The instance was associated with EC2-SG.

Figure 7. Blue-Server and Green-Server in EC2.

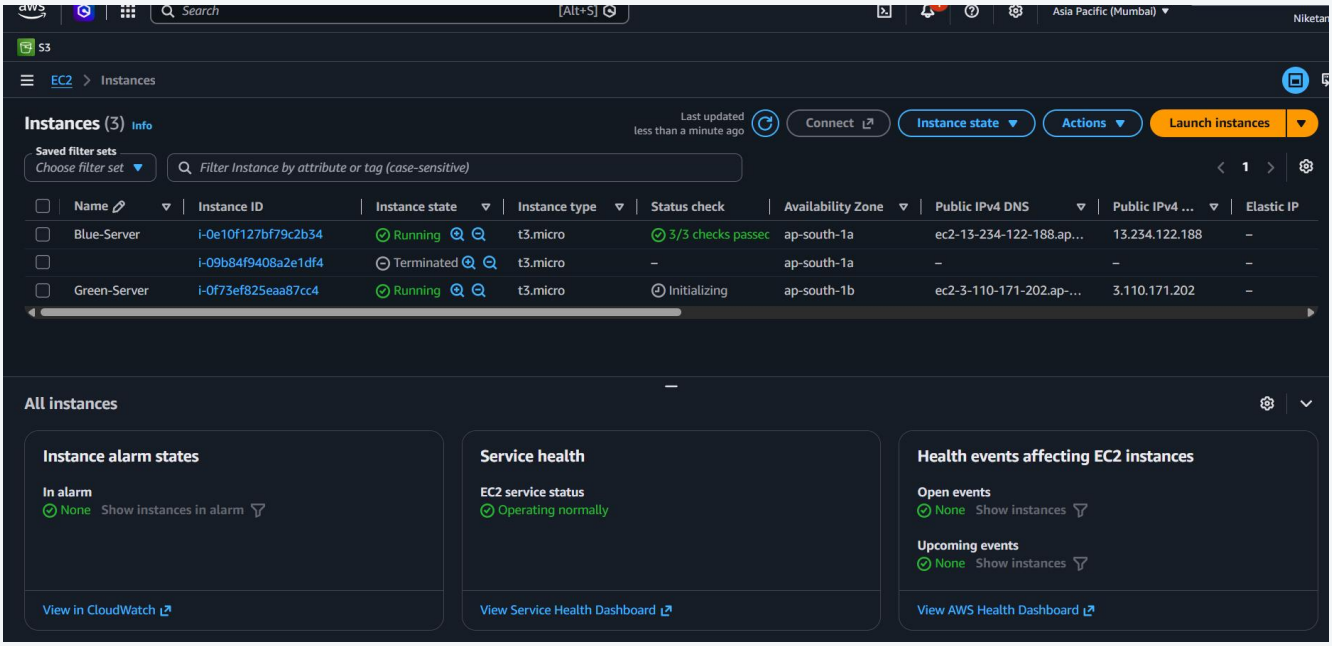


Figure 8. Blue Environment website.

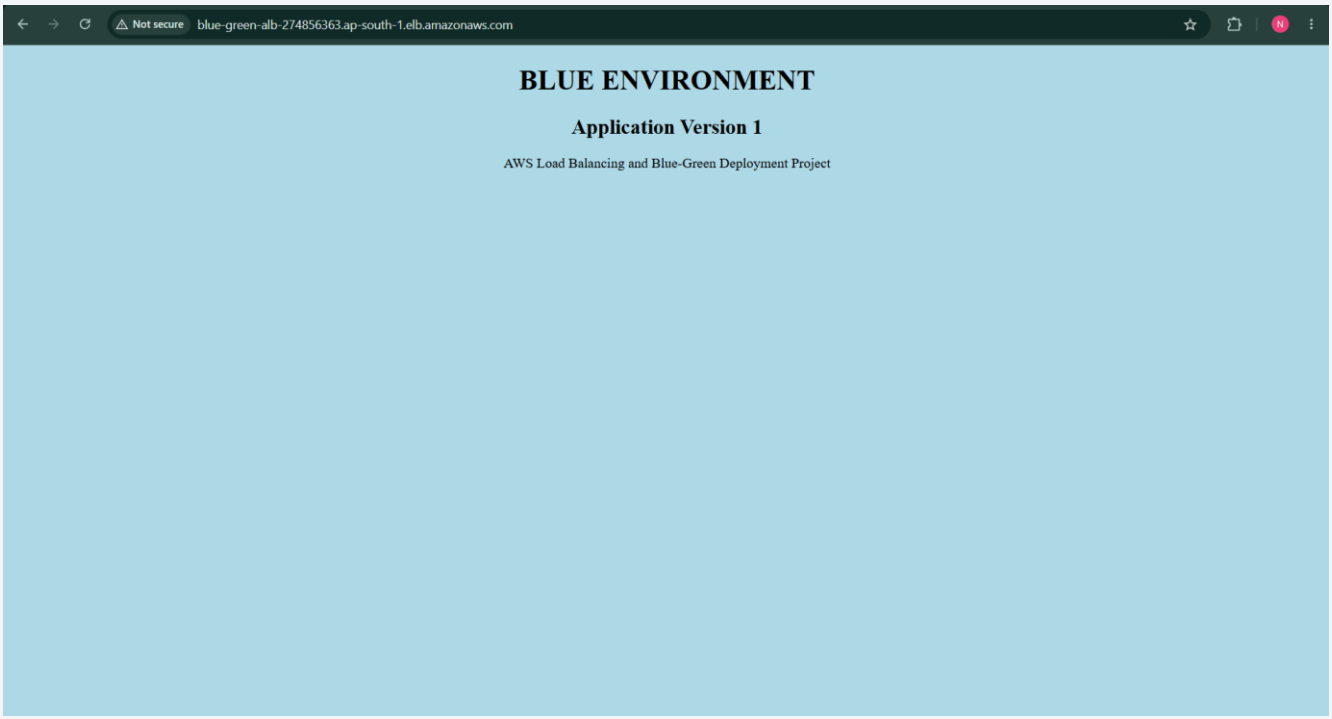


Figure 9. Green Environment website.



Step 5 — Create Target Groups

Two target groups were created with target type Instances, protocol HTTP, port 80, and health check path /. Blue-TG registered Blue-Server, while Green-TG registered Green-Server. Each target group kept its environment separate so it could be selected independently by the ALB listener.

Figure 10. Blue-TG target health.

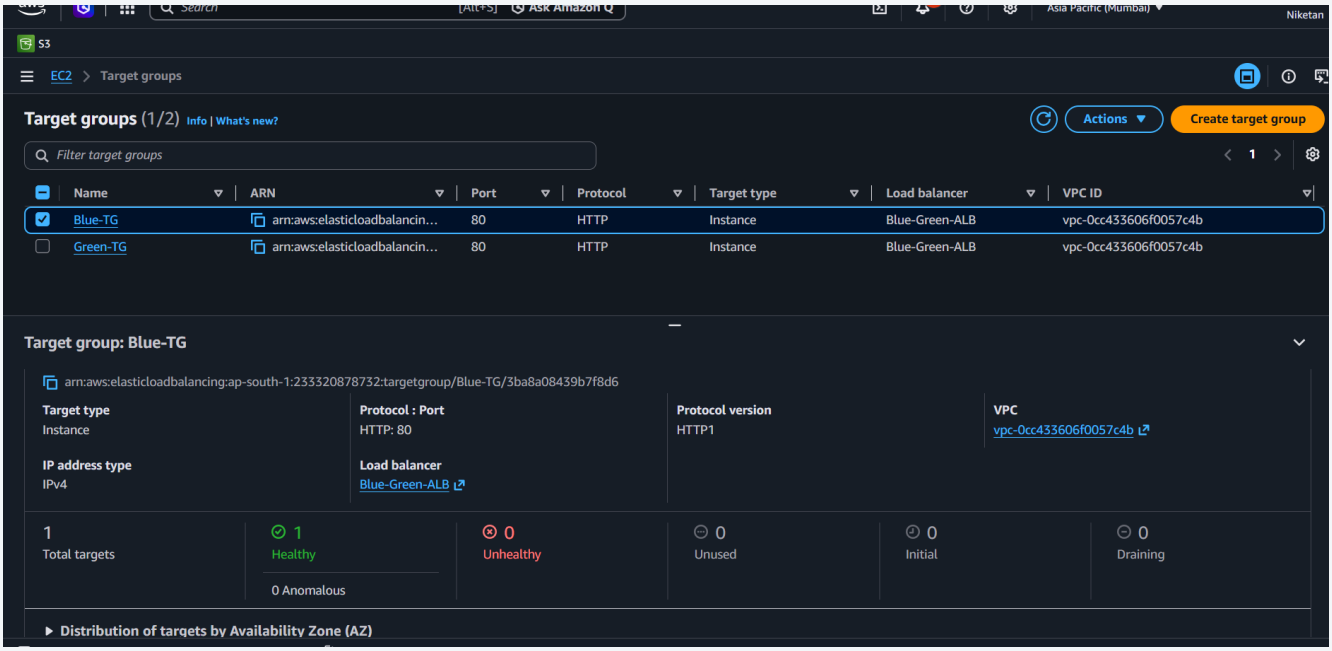
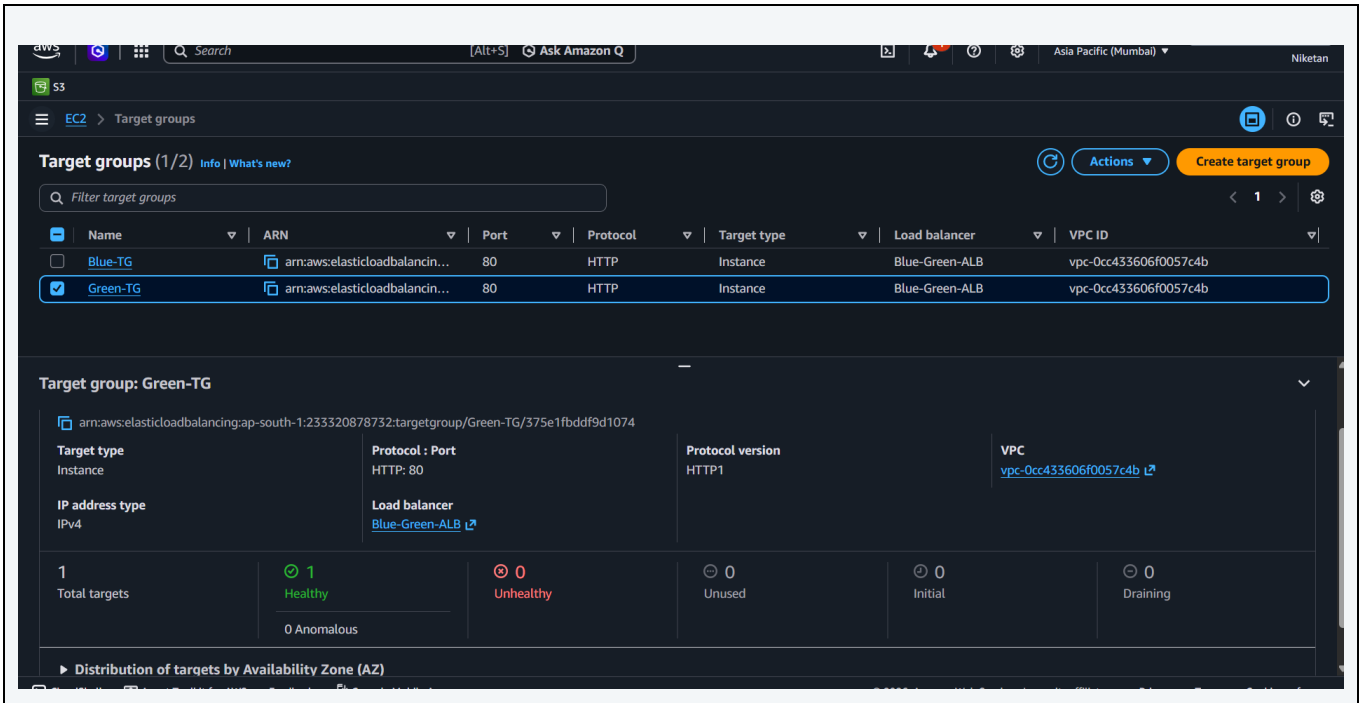


Figure 11. Green-TG target health.



Step 6 — Create the Application Load Balancer

An internet-facing IPv4 Application Load Balancer named BlueGreen-ALB was created. It was mapped to the two public subnets in different Availability Zones and associated with ALB-SG. An HTTP listener on port 80 was configured with Blue-TG as the initial default forwarding target group.

Figure 12. BlueGreen-ALB details and Active state.

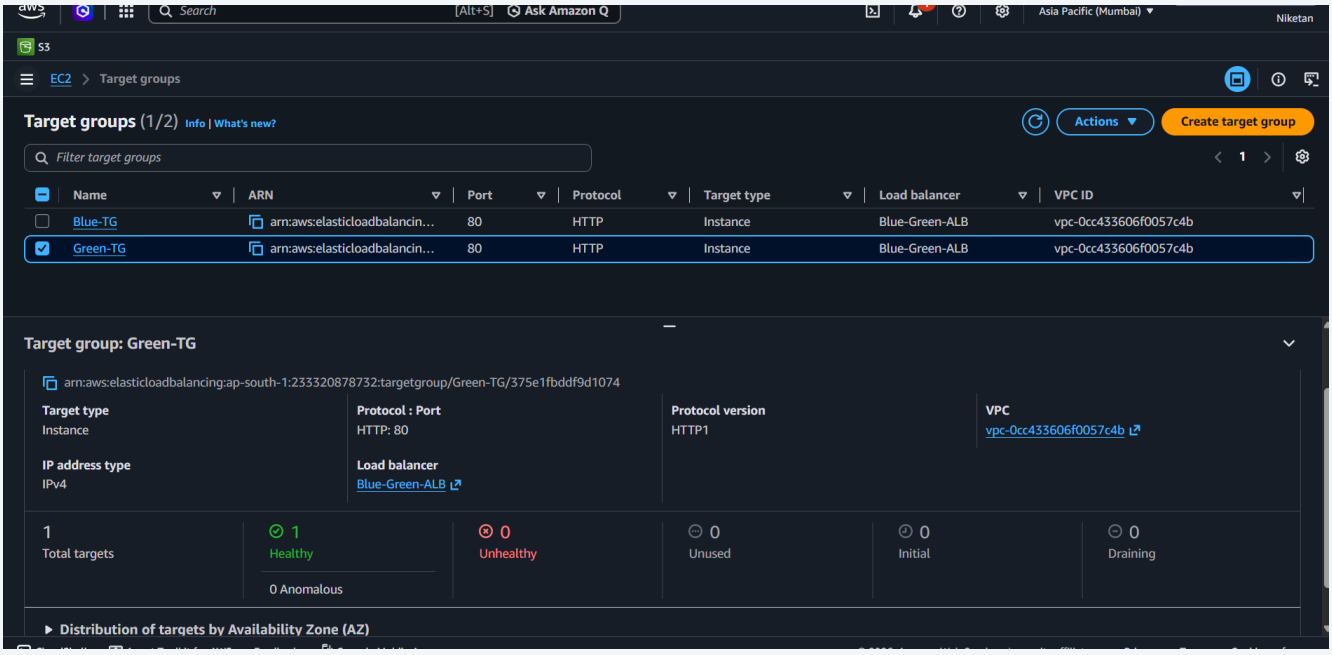
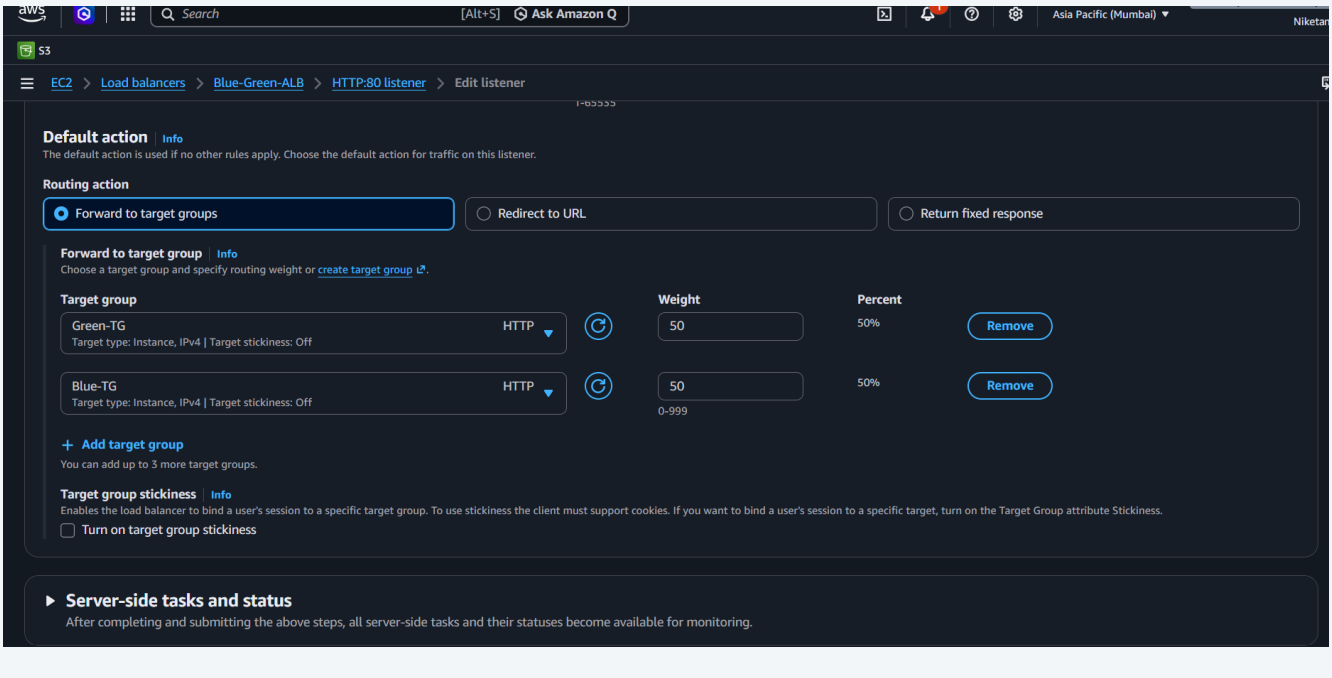


Figure 13. HTTP listener forwarding to Blue-TG.



Step 7 — Verify Target Health and Access the Application

The Targets tab for both target groups was checked. Blue-Server and Green-Server reached Healthy status, confirming that the configured HTTP health checks were passing. The application was accessed using the ALB DNS name rather than using an EC2 public IP address.

5. Blue-Green Traffic Switching

Blue-Green deployment maintains two application environments. Blue represents Version 1 and Green represents Version 2. The ALB listener's default forwarding action determines which environment receives traffic.

Deployment stage	Listener forwarding action	Expected result
Initial deployment	Blue-TG	The ALB DNS name displays BLUE ENVIRONMENT — Application Version 1.
Green deployment / cutover	Green-TG	The same ALB DNS name displays GREEN ENVIRONMENT — Application Version 2.
Rollback	Blue-TG	Traffic returns to the previous Blue version.

To switch environments, the default listener action was changed from Blue-TG to Green-TG. To roll back, the action was changed back to Blue-TG. The ALB DNS name remained unchanged; only the target group receiving traffic changed.

Figure 14. Listener forwarding to Green-TG.

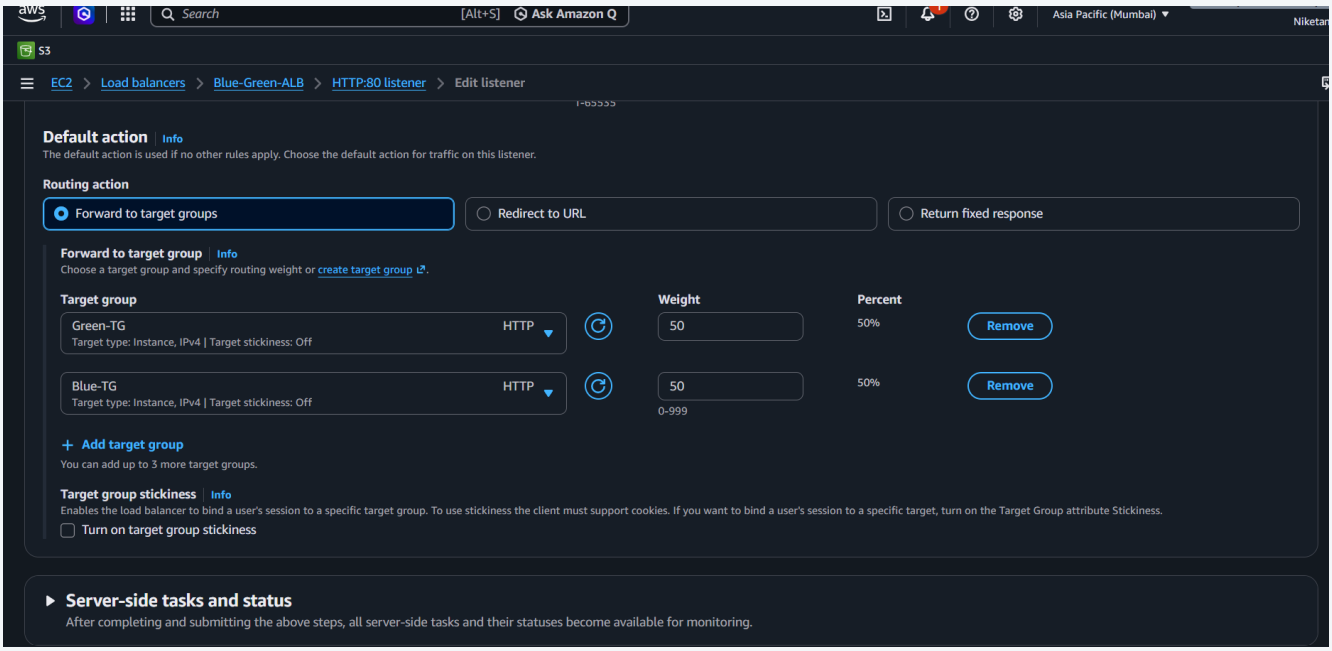
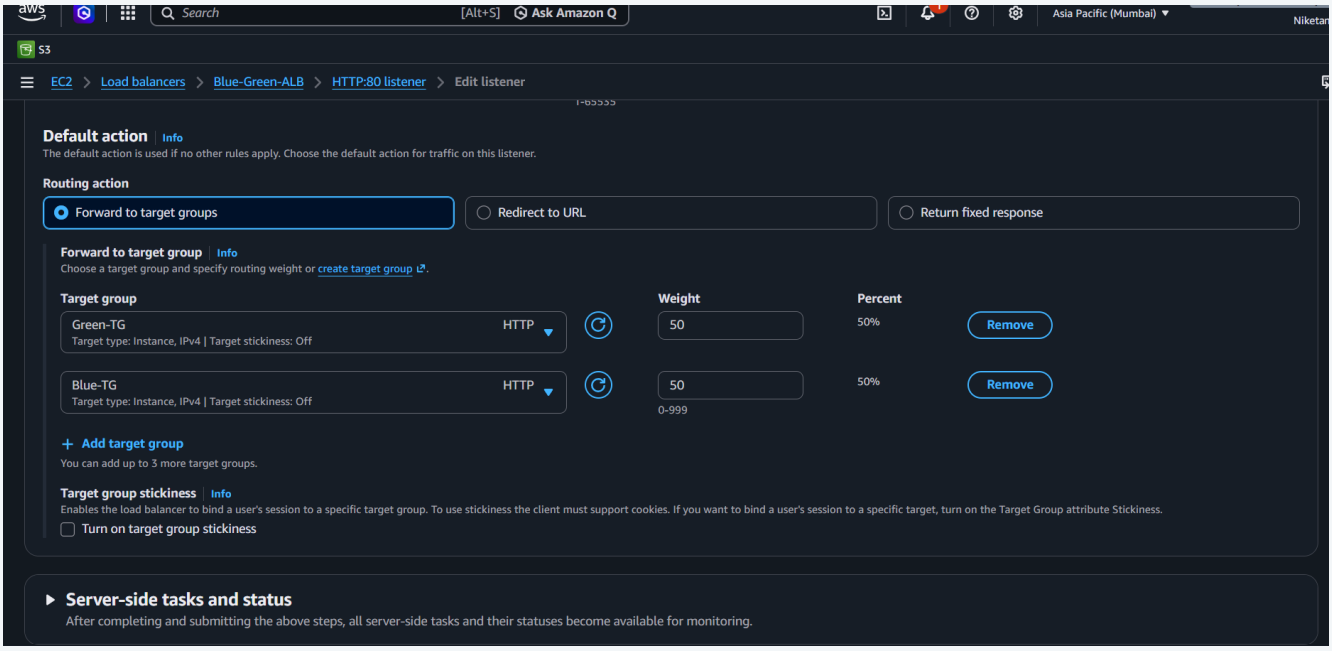


Figure 15. Rollback to Blue-TG.



6. Weighted Forwarding: 50% Blue / 50% Green

The ALB listener was also configured to forward requests to both target groups using a weighted forwarding action. Blue-TG was assigned weight 50 and Green-TG was assigned weight 50 in the same forwarding action.

Target group	Configured weight
Blue-TG	50
Green-TG	50

The weights represent relative routing proportions, not a guarantee that every pair of requests will contain one Blue response and one Green response. Short refresh tests may show the same environment several times in a row. With sufficient requests, the observed distribution may approach the configured 50/50 proportion. Target group stickiness, if enabled, can also affect observed behavior.

Figure 16. ALB weighted forwarding configuration.

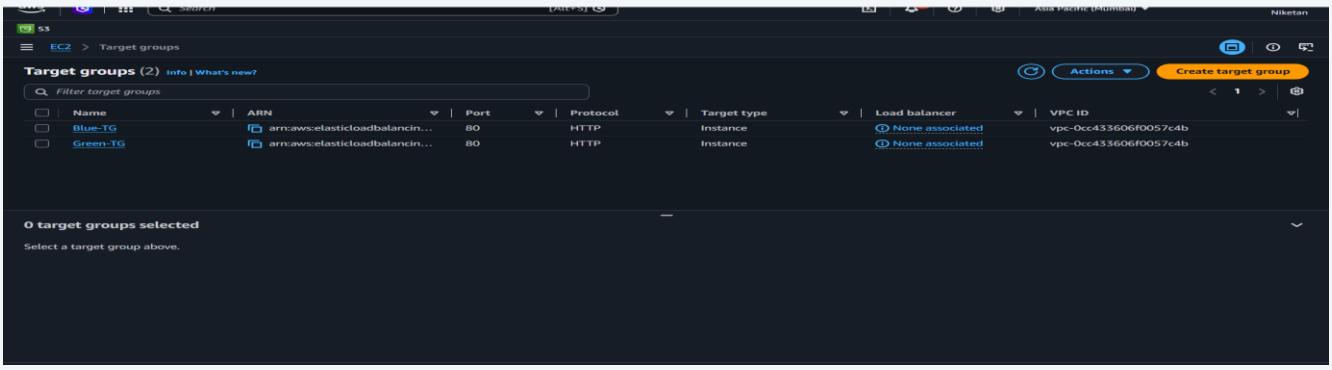


Figure 17. Browser output from Blue during weighted routing.

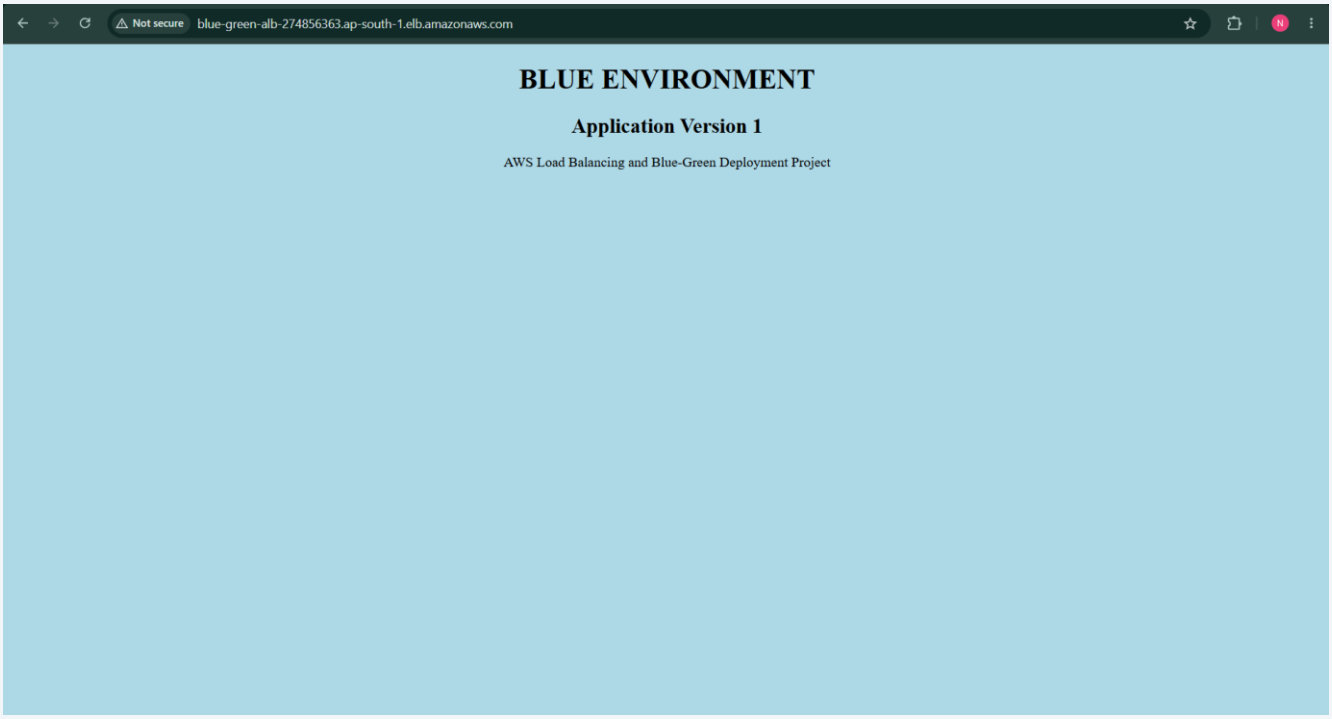
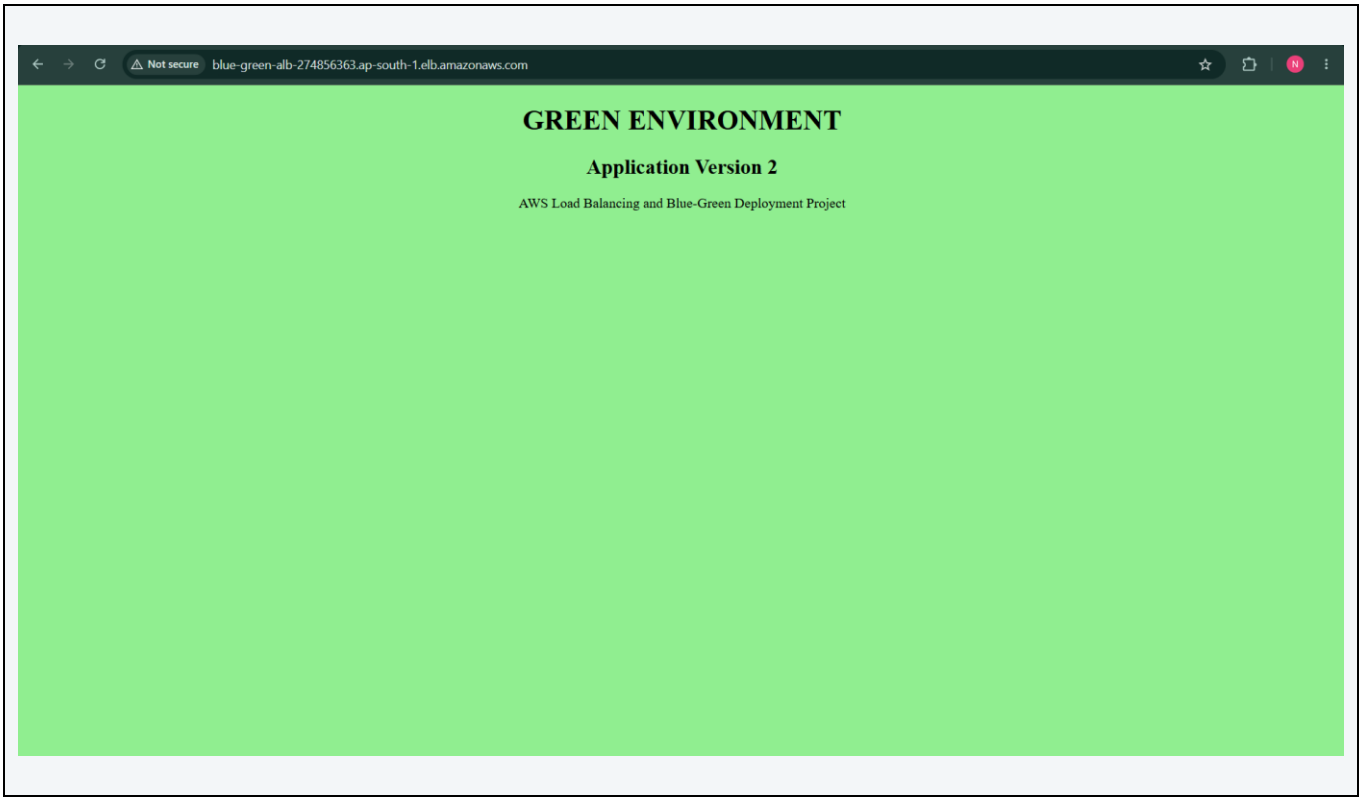


Figure 18. Browser output from Green during weighted routing.



7. Troubleshooting

Symptom	Possible cause / check
Target shows Unused	Check whether the target group is referenced by a listener rule or forwarding action, and whether the rule is active.
Target shows Unhealthy	Check Apache service status, port 80, health-check path /, target group port, VPC association, and EC2-SG allowing HTTP from ALB-SG.
ALB DNS name does not load	Check ALB state, listener port 80, ALB-SG inbound HTTP rule, subnet mappings, and Internet Gateway route.
Green page returns 404 on /green	ALB forwards the original path unless a URL rewrite/redirect is configured. Apache must handle that path, or use the default listener action to switch between Blue-TG and Green-TG.
Only one environment appears during 50/50 test	Verify both target groups are in the same weighted forwarding action and both targets are Healthy. A 50/50 weight does not guarantee alternating responses on each refresh.

8. Security and Cost Considerations

- Only the ALB security group was intended to accept public HTTP traffic. EC2-SG allowed HTTP from ALB-SG.
- For a production system, prefer private EC2 subnets and expose only the ALB. This lab used public subnets to keep the learning setup straightforward.
- HTTP port 80 is unencrypted. A production deployment should consider HTTPS using an appropriate TLS certificate and listener configuration.
- Avoid opening SSH (port 22) to 0.0.0.0/0. Restrict administrative access or use managed access methods.

- AWS resources can incur charges while running. After collecting project evidence, the ALB, target groups, EC2 instances, security groups, and project VPC were deleted.
- A final billing review should include EBS volumes, Elastic IP addresses, snapshots, and any other resources that may exist independently of the deleted VPC.

9. Resource Cleanup

After screenshots and project evidence were collected, the project resources were cleaned up. The Application Load Balancer was deleted first, followed by both target groups, both EC2 instances, the two custom security groups, and the custom VPC. A final review of independently retained resources and AWS billing is recommended.

Resource	Cleanup action
BlueGreen-ALB	Deleted
Blue-TG and Green-TG	Deleted
Blue-Server and Green-Server	Terminated
ALB-SG and EC2-SG	Deleted
lb-bluegreen-vpc	Deleted

10. Conclusion

This project demonstrated how an AWS Application Load Balancer routes HTTP requests to EC2 instances through target groups. Two separate Apache-hosted application versions were deployed and monitored with target health checks. The ALB listener was used to switch traffic between Blue and Green environments, perform a rollback, and configure weighted forwarding with a 50/50 split. The project provided practical experience with AWS networking, security groups, EC2, target groups, health checks, load balancing, and manual Blue-Green deployment.

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